

Mugahed Izzeldin

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PROFESSIONAL SUMMARY

PhD Candidate in Computer Science conducting systems research in cloud computing, mobile core networks, and reproducible infrastructure. Author of peer-reviewed and under-review publications spanning 5G core performance, reproducible systems, mobile roaming, and trusted execution environments. Owner and maintainer of NixCore, an open-source reproducible mobile network core framework. Experienced in designing and evaluating systems end-to-end: from implementation and experimentation through publication, documentation, and cross-layer debugging. Strong foundation in Linux, Nix/NixOS, Kubernetes, distributed systems, and production networking across 2G through 5G.

SKILLS

Systems: Linux, Nix/NixOS, reproducible infrastructure, distributed systems, mobile network core, Raft consensus, key-value stores

Cloud & Infrastructure: Kubernetes, Docker, Helm, Ansible, multi-tenant cloud operations

Networking: 5G Core (5GC), 4G Core (EPC), MP-BGP, BGP, OSPF, MPLS, L2/L3 switching and routing, TCP/IP, QoS, IPsec, VPNs

Programming: C, Go, Python, Shplait, JavaScript, HTML, CSS.

Tools & Operations: Git, CI/CD, monitoring, production troubleshooting, technical documentation

Databases: PostgreSQL, MySQL, MongoDB

EXPERIENCE

Graduate Research Assistant

Aug 2021 – Present

School of Computing, University of Utah

- Conducting systems research in the Flux Research Group under Prof. Robert Ricci, focusing on cloud computing, mobile networks, reproducible infrastructure, and networked systems.
- I own a research software end-to-end, from design and implementation through experimentation, debugging, documentation, and publication.
- Designing and evaluating mobile core research infrastructure, including reproducible deployments using DevOps and cloud-native tooling such as Kubernetes, Nix/NixOS, and containerized network components.
- Supporting CloudLab/Powder research and infrastructure in a large-scale experimental cloud used by distributed systems and networking researchers.

5G Core Network Research Engineer

May 2022 – Aug 2022

Celona Inc.

- Contributed research and engineering expertise to Celona's private 5G network solutions.
- Investigated commercial 5G core behavior in private-network deployments, conducted performance evaluation, and published the findings in a peer-reviewed publication.
- Evaluated system behavior and deployment tradeoffs across mobile core, networking, and cloud infrastructure layers.

Core Network Engineer

Mar 2021 – Aug 2021

Atlas Telecommunication

- Managed LTE core, IP networking, and transmission operations for a private LTE network, with end-to-end responsibility across core, transport, and access-network interfaces.
- Diagnosed and resolved production issues across Packet core, routing, and transmission layers, coordinating resolutions and service restoration across technical teams.

Core & MPBN Back-Office Engineer

Oct 2017 – Mar 2021

Huawei Technologies Sudan

- Operated and supported mobile Packet core and IP backbone infrastructure for service-provider networks, spanning core nodes, routers, firewalls, and customer-facing services.
- Configured, tested, launched, and troubleshot roaming and integrations between Packet core elements and 2G/3G/4G radio access networks.
- Analyzed Packet core and IP backbone KPIs to identify and resolve service-impacting performance and availability issues.
- Implemented network solutions, resolved production incidents, and expanded network capacity to improve service availability.
- Designed and developed internal tools for Packet core documentation, reporting, and analysis.

Core Network Planning and Optimization Engineer

Feb 2017 – Oct 2017

MTN Sudan

- Designed, dimensioned, and documented Packet core and IP core network changes for a mobile service provider environment.
- Analyzed service KPIs and coordinated network optimization to maintain performance, reliability, and service quality.

- Integrated new commercial services and data bundles into Packet core systems, defining change requests and operational procedures for production deployment.

PROJECTS

NixCore Framework nixcore.org

Project Owner

- Designed and currently maintain NixCore, a reproducible mobile network core framework for deploying, configuring, and evaluating 4G/5G core components using Nix/NixOS, Kubernetes, and declarative infrastructure.
- Built the framework to support repeatable research environments, reducing deployment drift, and making Packet core experimentation easier to reproduce and debug.
- I lead the project direction, implementation, documentation, and ongoing maintenance as an open-source research framework.
- Developed NixCore as the foundation for the research paper, "NixCore: Toward a Reproducible Mobile Network Core."

pomo Nix Package

search.nixos.org/packages?channel=unstable&query=pomo#show=pomo

Maintainer

- Maintain the pomo package in Nixpkgs, contributing to open-source packaging, reproducible builds, version updates, and package quality for Nix/NixOS users.

Missing Semester Arabic Website missing-semester-ar.github.io

Maintainer

- I maintain the Arabic version of MIT's Missing Semester course website, supporting accessible technical education for Arabic-speaking students and engineers.

Raft-Based Key-Value Store

- Implemented a distributed key-value store based on the Raft consensus algorithm, covering replicated state, leader-based coordination, and fault-tolerant system behavior.

Production Network Infrastructure Projects

- Designed and implemented production network services across ISP and Packet core environments, including DNS, RADIUS authentication, internet filtering, IPv6 designs, and GRX roaming architecture.
- Contributed to MTN Sudan's IP core migration between vendors and transition toward cloud-native infrastructure, including edge cloud integration, hardware expansion, and VNF scale-out for Packet core systems.

EDUCATION

University of Utah

PhD in Computer Science.

Expected Jun 2027

Relevant Coursework: Systems, Cloud Computing, Reproducible Infrastructure, Mobile Networks

Sudan University of Science and Technology

BSc. Hon Electronics Engineering (Honors).

SELECTED PUBLICATIONS

- "RaaS: How to Fix Mobile Roaming" Mugahed Izzeldin, Ryan W. West, Mary Ogunmola, and Robert Ricci. Manuscript under review, 2026.
- "NixCore: Toward a Reproducible Mobile Network Core" Mugahed Izzeldin and Robert Ricci. Manuscript under review, 2026.
- "Confidential5G: Trusted Execution of the 5G Core on Untrusted Clouds" Mary Ogunmola, Mugahed Izzeldin, Robert Ricci. Manuscript under review, 2026.
- "Analysing the performance of commercial 5GC in private network deployment" Mugahed Izzeldin, Lyutianyang Zhang, Vanlin Sathya, Mohit Goyal, and Mehmet Yavuz. International Journal of Mobile Network Design and Innovation, 2024.
- "Approaches for SMS encryption and user accounts verification" M. I. O. Hajahmed, K. E. M. Osman, and O. T. M. Ali. ICCCEE20 Conference, 2021.
- "Implementation of SDN in a Campus NAC Use Case" M. I. O. Hajahmed, M. Adil, M. Omar, M. Khalid. B.Sc. (Honors) degree project.

PROFESSIONAL DEVELOPMENT & SERVICE

- Conference attendance: ACM CCS 2024; KubeCon + CloudNativeCon North America 2024.
- Certifications: Huawei HCIA Routing and Switching, CCNP ROUTE, CCNA, IPv6 Workshop by AFRINIC.
- Coursework: SQL, Python, data analysis, and web development.
- Volunteer service: IEEE Sudan Subsection Media Coordinator, IEEEExtreme proctor, IEEE SUST branch election organizer, technical seminar presenter.